



Design Narrative

Replace Vault and Fuel Tank, B3202

Project Number: WWYK230053

Work Order: 1134513

Tinker Air Force Base, Oklahoma

June 2, 2025

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1. Overview

The project at Building 3202 includes upgrades to civil, architectural masonry repairs and new roof, structural, mechanical, and electrical systems.

Work involves removing the underground fuel vault, replacing the 1,400 SF roof, and installing a new above-ground diesel tank inside the building. The design also adds new ventilation and fuel piping. Electrical updates include LED lighting with battery backup, new panels, and fuel system connections. The structural scope includes supporting the new MEP installations and repairing the corroded steel hoist structure. All improvements are designed to meet current codes and Tinker AFB standards.

2. Applicable Codes and Standards

2.1 General

- IBC International Building Code (as amended by UFC 1-200-01)
- IEBC International Existing Building Code
- NFPA 101 Life Safety Code
- TAFB Facility Standard
- UFC 1-200-01 DoD Building Code
- UFC 4-010-01 DoD Minimum Antiterrorism Standards for Buildings

2.2 Civil

- BASH Manual
- UFC 3-201-01 Civil Engineering
- UFC 3-250-01 Pavement Design for Roads and Parking Areas
- UFC 4-022-03 Security Fences and Gates

2.3 Architectural

- ABA Architectural Barriers Act Standards
- UFAS Uniform Federal Accessibility Standards
- UFC 3-101-01 Architecture
- UFC 3-110-03 Roofing
- UFC 3-190-06 Protective Coatings and Paints
- UFC 3-201-02 Landscape Architecture

2.4 Structural

- ACI 318 Building Code Requirements for Structural Concrete and Commentary
- AISC 360 Steel Construction Manual
- ASCE 7 Minimum Design Loads for Building and other Structures
- UFC 3-301-01 Structural Engineering

2.5 Mechanical / Plumbing

- ANSI/ASHRAE/ACCA Standard 183 Peak Cooling and Heating Load Calculations in Buildings Except Low-Rise Residential Buildings
- ASHRAE 55 Thermal Environmental Conditions for Human Occupancy
- ASHRAE 62.1 Ventilation for Acceptable Indoor Air Quality
- ASHRAE 90.1 Energy Standard for Buildings Except Low-Rise Residential Buildings
- IECC International Energy Conservation Code
- IMC International Mechanical Code (as amended by UFC 3-410-01)
- IPC International Plumbing Code (as amended by UFC 3-420-01)
- NFPA 20 Standard for the Installation of Stationary Pumps for Fire Protection
- NFPA 25 Standard for the Inspection, Testing, and Maintenance of Water-Based Fire Protection Systems
- NFPA 30 Flammable and Combustible Liquids Code
- NFPA 31 Standard for the Installation of Oil-burning Equipment
- NFPA 37 Standard for the Installation and Use of Stationary Combustion Engines and Gas Turbines
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- TAFB Mechanical Standard
- UFC 1-200-02 High Performance and Sustainable Building Requirements
- UFC 3-400-02 Design: Engineering Weather Data
- UFC 3-401-01 Mechanical Engineering
- UFC 3-410-01 Heating, Ventilating, and Air Conditioning Systems
- UFC 3-410-02 Direct Digital Control for HVAC and Other Building Control Systems
- UFC 3-410-04 Industrial Ventilation
- UFC 3-420-01 Plumbing Systems
- UFC 3-430-01FA Heating and Cooling Distribution Systems
- UFC 3-450-01 Noise and Vibration Control

2.6 Electrical

- NFPA 70 National Electrical Code
- TAFB Electrical Standard
- UFC 3-501-01 Electrical Engineering
- UFC 3-520-01 Interior Electrical Systems
- UFC 3-530-01 Interior and Exterior Lighting Systems and Controls
- UFC 3-540-01 Engine-Driven Generator Systems for Prime and Standby Power Applications
- AFMAN 32-1065 Grounding and Electrical Systems

2.7 Fire Protection

- NFPA 1 Fire Code
- NFPA 13 Standard for the Installation of Sprinklers
- NFPA 20 Standard for the Installation of Stationary Pumps for Fire Protection
- NFPA 70 National Electric Code
- NFPA 72 National Fire Alarm Code
- NFPA 241 Standard for Safeguarding Construction, Alteration, and Demolition Operations
- TAFB Fire Protection Standard
- DAFI 32-2001 Fire and Emergency Services (F&ES) Program
- UFC 3-600-01 Fire Protection Engineering for Facilities
- UFC 3-601-02 Operation and Maintenance: Inspection, Testing and Maintenance of Fire Protection Systems
- UFC 4-021-01 Design and O&M: Mass Notification Systems

3. Civil Design Narrative

The civil requirements for the project include (but are not limited to):

- Remove existing underground concrete vault and fuel tank.
- Remove existing sidewalk and replace with new concrete truck driveway.
- Repair or replace existing fence.

3.1 General Design Parameters and Assumptions

Work will be performed in accordance with relevant sections for demolition and pavement replacement in UFC 3-201-01 Civil Engineering.

3.2 Civil Demolition

Conduct tests of the soil by an approved laboratory to confirm no hazmat material exists in the immediate vicinity of the underground concrete vault.

Excavate all material in and around the vault to the extent needed to remove all hazardous material to the satisfaction of the COR.

Dispose of any hazardous materials off base to an approved disposal site. Dispose tickets to the approved site must be furnished to the COR for all loads.

Remove the existing sidewalk in front of Building 3202 to prepare for the future installation of the concrete drive pad.

Remove the existing guardrail east of building 3202.

Demolish existing concrete lid above existing vault.

Demolish existing fuel tank within the underground vault.

Demolish the entire underground concrete vault.

Excavating soil around existing vault is necessary to facilitate the removal of existing concrete walls and concrete base of storage tank vault.

Backfill void left by removal of underground storage tank vault and provide grading to match existing drainage patterns. Install new sod at the disturbed areas.

Remove existing sidewalk on west side of B3202 and replace with new concrete pad approx. 37' x 12' x 8" thick concrete drive.

The designated staging area for this project is located in the parking lot east of Building 3202 and the water tank. This area will be used for the storage of materials, equipment, and tools necessary for construction activities. Contractors are required to maintain the staging area in an organized and safe condition throughout the duration of the project.

Note: All contractor privately owned vehicles (POVs) must be parked on the west side of Building 3202, just west of the fence.

3.3 Civil Construction

Backfill the entire excavation with approved select fill compacted to 95% Modified Proctor Density in 6-inch lifts to 12-inches below top of the existing grade.

Excavate the area to facilitate proper drainage. Install sod grass over the area to restore its surface and ensure stabilization.

Construct an 8-inch reinforced concrete driveway for fuel truck access west of building 3202.

Repair the security fence on the south side of building 3202.

Repair the 7-foot chain link fence with 3-strand barbed wire around Building 3202 as needed.

Remove the existing gate on the east side of the fence.

Install a new 14-foot-wide double swing gate on the west side of the building to provide fuel truck access.

Repair the existing security fence along the south side of the building to maintain site security and controlled access.

3.3.4 Perimeter Fencing

7-foot chain link fence with 3-strand barbed wire will be used for all fencing modifications around Building 3202. The scope includes removal of the existing gate on the east side of the fence, installation of a new 14-foot-wide double swing gate on the west side of the building to provide fuel truck access, and repair of the existing security fence along the south side of the building to maintain site security and controlled access

3.3.5 Grading and Drainage

Excavate the area east of building 3202 to facilitate proper drainage. All vegetation disturbed as part of this construction, including staging areas will be replanted with Bermuda grass meeting BASH requirements and certified red ant free.

3.3.6 Erosion Control

An erosion control plan is provided. The plan utilizes Best Management Practices (BMP) to control the runoff of silt and sediment from the site during rainfall events and reduce the amount of sediment leaving the site resulting from construction activities.

4. Architectural Design Narrative

The architectural scope addresses water infiltration damage to the exterior and interior masonry at Building 3202.

4.1 Architectural Demolition

Remove exterior door and frame.

Remove the existing built-up roofing system down to the existing deck.

Remove all the existing wood blocking, rigid insulation, and recovery board.

Remove all pitch pans and pipe penetration flashings.

Remove the existing roof drain.

4.2 Architectural Construction

Replace exterior door, frame and hardware with new metal door/frame, lever handle, lockable. Flash and seal frame and door to be weather-tight.

Furnish and install new full adhered white EPDM roof membrane system applied over mechanically fastened insulation and recovery board substrate, Roof system to include EPDM sheet, Rigid insulation, Tapered cricket/saddle insulation and High-Density Polyisocyanurate cover board.

Furnish and install new 24-gauge metal. through wall scuppers and pictures frame. Provide a polyvinylidene fluoride (PVF2) coating (similar but not limited to Kynar 500).

Raise parapet walls to achieve a minimum membrane flashing height of 8" from top of the finished roof membrane to top of the flashing membrane.

Remove the existing roof drain and install a new primary and secondary drainage system.

The required new 90 mil EPDM roofing system will have a 30-year warranty, and the roof system installer must warrant for a period of two (2) years.

Remove the efflorescence from the interior and exterior faces of the brick and concrete stone band using a strong brush and diluted muriatic acid mixture or similar. Do not power wash with pressurized water to prevent further water intrusion. After drying, apply baking soda to neutralize the acid and prevent any additional damage to the masonry. Clean wall surfaces thoroughly.

Provide new backer rod and sealant around all wall openings, doors, penetrations, and cracks.

Repoint all exterior wall mortar joints. Replace 8"x32" portion of damage concrete stone band in 2 places. Existing cast stone cap will need to be removed and reinstalled on top of metal parapet cap. Provide 22-gauge internal drainage system with no new penetrations to the exterior wall.

The Contractor shall comply with all stipulations and requirements outline in any executed Memorandum of Agreement (MOA) developed in coordination with the Oklahoma State Historic Preservation Office (SHPO). Work shall not proceed in areas subject to MOA Stipulations until all application requirements have been satisfied.

5. Structural Design Narrative

The structural scope involves aiding new MEP work in the building and repairing corroded steel hoist structure.

5.1 General Design Parameters and Assumptions

5.1.1 Concrete and Reinforcing Steel

Structural Cast-In-Place Concrete, Unless Otherwise Indicated: 4000 psi
Concrete for Paving, Sidewalks, slab-on-grade and Other Flatwork: 4000 psi
Cement: Type II or I/II for cast-in-place concrete.
Reinforcing Steel (Deformed Bars): ASTM A615, Grade 60.
Reinforcing Steel to Be Welded (Deformed Bars): ASTM A706, Grade 60.
Anchor Bolts: ASTM F1554, Gr 36, Galvanized.

5.1.2 Reinforced Concrete Design

Design of Slab-On-Ground: Slabs-on-ground shall be designed in accordance with ACI 360R and ACI 318.

5.1.3 Structural Steel

W and WT Shapes: ASTM A992 or ASTM A572, Grade 50, with special requirements per AISC Technical Bulletin #3.
Other Rolled Sections and Plates: ASTM A36 or ASTM A572, Grade 50.
Square or Rectangular Tubes (HSS): ASTM A500, Grade B.
High Strength Bolts: ASTM A325, with compressible washer type direct tension indicators or twist of type tension control bolts.
Anchor Bolts: ASTM F593, Type 304.

5.1.4 Steel Design

Structural steel design will be performed in accordance with the requirements of the American Institute of *Manual Steel Construction*, AISC 360.

5.1.5 Structural Loads

5.1.5.1 Dead Loads

Dead loads will be considered resulting from fixed construction, materials, and equipment. Dead loads will be determined and applied in accordance with ASCE 7, and other pertinent standards and vendor-supplied information.

5.1.5.2 Live Loads

Live loads are those loads produced by the use and occupancy of the building or structure, and do not include loads such as wind/seismic loads, rain loads, flood loads, or dead loads. Minimum uniformly distributed live loads will be as required by ASCE 7.

5.1.5.3 Wind Loads

General: Wind loads for structural design of main wind force resisting systems and components and cladding shall be determined and applied in accordance with ASCE 7, Directional procedure.

Wind Speed: The minimum 3-second gust design wind speed shall be determined from Figure 26.5 of ASCE 7. A basic wind speed of 120 mph will be appropriate for the project location.

Exposure Category: Building exposure classification C, as defined by ASCE 7, shall be used for this building.

Risk Category: Building risk category IV, as defined by ASCE 7, shall be used for this building.

5.1.5.4 Load Combinations

Load combinations using strength design and service-level combinations shall be as defined in ASCE 7.

5.1.5.5 Seismic and Snow Loads

Seismic loads shall be determined in accordance with IBC. The structure will be designed with Seismic Site Class of "D" based on the seismic parameters defined in IBC.

Snow loads shall be determined in accordance with IBC. The ground snow load shall be 22 pounds per square foot.

5.2 Structural Demolition

5.2.1 Mechanical and Electrical

Demolish existing structural components supporting MEP as required to accommodate new MEP work.

5.2.2 Steel Hoist

Demolish existing corroded base as required for repair.

5.3 Structural Construction

5.3.1 Equipment Pad

Design and construct a new concrete equipment pad to support the new fuel tank.

5.3.2 Steel Hoist

Design and construct new base repair for corroded structure.

5.3.3 Concrete Repair

Repair concrete damage throughout the structure as required. There is currently a large crack at the entry slab.

5.3.4 Masonry Walls

Repoint the exterior and interior masonry walls.

Seal the walls as required to prevent future water infiltration.

6. Mechanical Design Narrative

The underground fuel tank at B3202 does not meet current regulations from the State of Oklahoma. The underground vaulted tank will be demolished, and the new tank will be installed above ground inside B3202. The mechanical scope on this project has been revised since the PAA was performed; since it was discovered that the original proposed design would not meet NFPA 20 (Section 11.4.3.1 - Diesel fuel supply tanks shall be located above ground in accordance with municipal or other ordinances and in accordance with requirements of the authority having jurisdiction and shall not be buried. 11.4.3.2 - In zones where freezing temperatures [32°F (0°C)] are possible, the fuel supply tanks shall be located in the pump room.) Also, with the adoption of the 2025 version of NFPA 20 the required size of the new fuel tanks has been increased from 8 hours to 12 hours of fire pump engine run time.

6.1 General Design Parameters and Assumptions

6.1.1 Elevation, Latitude/Longitude, Climate Zone

Elevation, latitude/longitude, and climate zone were taken from International Energy Conservation Code 2015. The weather data below was taken from UFC 3-400-02 and 2013 ASHRAE Fundamentals Chapter 14.

- Elevation: 1,260 ft.
- Latitude: 35.42 N
- Longitude: 97.38 W
- Climate Zone: 3
- Summer: 100°F DB/74°F WB
- Winter: 12°F

6.1.2 General Indoor Temperature/Relative Humidity Parameters

HVAC Design by Building Area				
Area Designation	Design Temperature (°F)		Relative Humidity (%)	
	Minimum Winter	Maximum Summer	Minimum Winter	Maximum Summer
Mechanical Room	50	N/A	N/A	N/A

6.2 Indoor Air Quality

Ventilation rates for all areas will be provided per ASHRAE 62.1-2016. Ventilation air will be provided by a combination of natural and mechanical ventilation being installed under this project.

6.3 Validation of Existing HVAC Systems

Validation of existing HVAC systems was accomplished by reviewing existing plans and field observation of the HVAC systems serving the areas within the building where renovation work is taking place.

6.4 Mechanical Demolition

Demolition includes but is not limited to the following:

6.4.1 General

All mechanical demolition work will require a licensed Oklahoma Corporation Commission (OCC) operator and a licensed OCC environmental consultant for soil sampling.

Prior to demolition of the existing underground tank, remove and return to the base fuel any fuel remains in the main tank and the underground fuel lines.

Test the soil to determine if the tank is leaking, determine if soil remediation is required, and inspect the underground pipe insulation for asbestos containing material (ACM).

6.4.2 B3202

Demolish the existing exhaust fan on the roof and all associated controls.

Demolish the two existing natural gas fired heating units' flues/vents and the diesel generator flue/vent from the roof back to the units to allow for roof replacement.

Demolish the existing door mounted louvers.

Demolish the existing roof drain and the associated piping.

Prior to demolition of the main vaulted tank disconnect the day tank from the main lines and the fire pump in an approved phasing schedule, so that no more than one fire pump is inoperable during demolition and construction.

Demolish the existing vaulted fuel storage tank and the underground vault once the new tank has been installed and connected to the existing pump.

Demolish all underground fuel piping from the tank up to the existing day tank and the pumps inside the building.

6.5 Mechanical Construction

6.5.1 General

All mechanical construction work will require a licensed Oklahoma Corporation Commission (OCC) operator.

Design and construction of the ventilation and fuel storage systems inside B3202 must be in accordance with NFPA 20, 30, 31, and 37.

6.5.2 B3202

Design, provide, and install a new ventilation system for the building in accordance with ASHRAE 62.1 and NFPA 37. The system shall be composed of an exhaust system that meets the ventilation requirements of NFPA 37 and a natural means to provide make-up air, such as through wall or door louvers. Louvers must be designed to resist wind-driven rain and be sized in accordance with ASHRAE 62.1. Design and furnish any necessary controls and connect equipment to one of the base wide DDC monitoring systems.

Design and install one new double-walled, grounded, above ground fuel tank inside B3202 to provide fuel for the fire pump. Tanks volume must be sized in accordance with the 2025 edition of NFPA 20. Double walled tanks interstitial space must be monitored for fuel leakage in accordance with NFPA 20, tanks must also be UL 142 rated and equipped with a UL 508 rated control panel.

Tanks shall have a fuel level indicator which activates when the tank capacity falls below two-thirds the tank's volume. New tanks must be compatible with the existing fuel alarm system installed in B1032. Fuel level sensor shall activate an audible alarm at the fill point when the level reaches 90% of the tank's capacity and an automated shutoff that stops the flow of fuel when 95% of the tank's capacity is reached when refueling to prevent overfilling.

Install new fuel connection lines from the tank to the fire pump in accordance with NFPA 20.

Install a new tank fill line to the West exterior wall of B3202 which connects to the tank for re-fueling purposes. The new tank fill line should be equipped with a filler neck and screw on cap that will accept a 1" auto shutoff diesel pump nozzle at the fill point. The new tank fill line should be equipped with a spill bucket at the fill points. BASIS OF DESIGN: Morrison Brothers 715 fill box with bottom connection.

Design, provide, and install a new roof drain system to facilitate water drainage from the roof.

6.5.3 Vibration Control

New exhaust fans will be provided with appropriate vibration isolators to minimize vibration into building structure.

6.5.4 Controls

Local control will be provided on new exhaust fans within the building. Metering of the gas, electricity, and domestic water usage for this building will not be done under this project as we are not modifying these incoming services under this project.

6.5.6 System Commissioning, Testing and Balancing

Provide commissioning of work designed/constructed under this project. Also provide to a firm certified by the Associated Air Balance Council (AABC) or National Environmental Balancing Bureau (NEBB) to perform the testing, adjusting and balancing procedures of air side and water side systems per the requirements of the AABC or NEBB Procedural Standards for Testing, Adjusting and Balancing of Environmental Systems.

7. Electrical Design Narrative

The electrical requirements for the project include (but are not limited to):

- Demolition of electrical equipment connections for the existing underground fuel tank.
- New electrical installation and connections for the new fuel tank above ground.
- Removal/replacement of existing lighting and power systems affected by water intrusion on the building.

7.1 General Design Parameters and Assumptions

The existing Building 3202 (Fire Pump Station) was constructed in 1943 and has 1359 square feet. The exterior consists of dark red field brick and concrete lintel characteristics of the original Douglas Aircraft Plant Historic District Construction. This facility was constructed as a part of the original building 3001 construction as a dedicated pump house for fire suppression water for a large portion of the industrial complex of TAFB.

The existing electrical system fed the main fire pump, additional sump pumping systems, the lighting system, and the electrical power and controls. There is an operational standby generator, automatic transfer switch, feeding the panelboards "EHP1" and "ELP1".

An Engineering Evaluation (WT 14043285), to verify the integrity of the building, and from previous building inspections (2008, 2021, 2023), concluded that there are severe damages that need to be corrected before the installation and/or replacement of the new equipment take place.

The existing panelboards were manufactured by Federal Pacific and Zinsco.

7.2 Electrical Demolition

7.2.1 Main Building

(1) Power

- (a) Contractor must verify and provide a design analysis of the existing loads, Short Circuit analysis, Protective Device Time-Current Coordination Study, Arc-Flash Analysis, Voltage Drop Analysis, Motor Starting/Flicker Analysis, prior the installation and connection of the new equipment for government approval.
 - i. Verify the latest As-Built panelboard schedules of the existing panels, if it's available.
 - ii. Identify each circuit, panels, and any other sensitive load and/or circuit. The proper tracing downstream circuits and marking unused circuits at spare.
 - iii. Verify the voltage/amperage capacity, and SCCR.
- (b) Disconnect and remove switchgear, MDP, panelboards, step-down transformers, and the section between the automatic transfer switch and the switchgear. The automatic transfer switch (ATS) and the emergency panelboard (EHP1) will remain in place but all damaged conduits and conductors must be replaced.
- (c) Disconnect and remove circuits and conduits for the main pump and sump pumps and any other associated component. All wiring will be removed back to the power source.
- (d) Disconnect and remove the step transformer mounted on the roof and overhead branch circuit. Remove the 30A disconnect of the Big Voice Pole.
- (e) Remove and dispose all empty and rusted conduits.

(2) Lighting

- (a) Disconnect and remove all suspended light fixtures, wall switches with conduits, j-boxes, supports and any other associated component. All wiring will be removed back to the power source.

7.2.2 Fuel Tank

- (1) Disconnect and remove the circuit that feeds the fuel tank sump pump. Remove conduits, conductors, supports, any other associated component.
- (2) Disconnect the alarm horn and /or annunciators.
- (3) Coordinate with other trades for additional demolition work within the renovated area.

7.3 Electrical Construction

7.3.1 Main Building

(1) General

- (a) New electrical equipment shall be installed in a professional and skillful manner per NEC Code and UFC specifications. The new location shall be indicated in the construction documents.
- (b) Contractors must verify all the field requirements and working space required per NEC.

(2) Power

- (a) New MDP panelboard, labeled as '3202-MDP-1'. Previously known as '3202-PNL-1'.
 - i. Provide a Surge Protective Device (SPD).
 - ii. All bolt-on breakers and additional spares will be required. 20% spare/space minimum.
 - iii. Contractor must provide new high-quality labels for all new equipment. New circuit directories updated on the panelboards involved.
 - iv. Reconnect the existing circuits.
- (b) Contractor will provide and install new service entrance conductors between the main disconnect and the new MDP.
- (c) New Branch Panelboard, labeled as '3202-PNL-1', known as 'ELP1'.
 - i. All bolt-on breakers and additional spares will be required. 20% spare/space minimum.
 - ii. Reconnect the existing circuits (lighting, fire alarm, receptacles).
 - iii. Contractor must provide new high-quality labels for all new equipment. New circuit directories updated on the panelboards involved.
- (d) Provide and install a new step-down transformer, labeled as '3202-XFMR-1'.
- (e) Provide and install new circuit to feed the Big Voice Pole.

(3) Lighting

- (a) Provide and install new LED weatherproof light fixtures.
 - i. Provide and install new lighting fixtures in the same location. Few of them with Integral battery backup for egress and access stairs lighting.

- ii. Provide new dual-tech occupancy sensor control and override with single pole light switches.
- iii. Provide and install new wallpack fixtures for exterior lighting and connected to the existing lighting circuit.
- iv. Remove and replace any photocell control system for the exterior lighting.

7.3.2 Fuel Tank

- (1) Provide and install the circuit required for the day-tank, and the associated connection with the control fuel system.
- (2) Install new Emergency Fuel Shut Off switch, alarm horn, level/leak sensors.
- (3) All wall penetrations must be properly sealed for water infiltration (for the demolished conduits and new installation).

7.4 Testing

Contractors will perform the proper tests to verify the functionality of the new equipment per ANSI/NETA ATS procedures.

8. Fire Protection / Life Safety Design Narrative

The existing fire pump house, B3202, will require approved fire protection laborers to provide the labor required for any modifications or relocation of fire alarm or fire suppression equipment and systems.

8.1 General Design Parameters and Assumptions

8.1.1 Jurisdictional Authority

The Contracting Officer and Base Fire Chief are the Authorities Having Jurisdiction (AHJs).

8.1.2 Fire Protection Approach

No new equipment is scheduled to be provided or installed. Existing fire protection equipment is planned to be relocated within the same room or space.

8.1.3 Occupancy Classification

The occupancy classification will remain unchanged as Utility, U-Occupancy.

8.2 Fire Protection Systems

8.2.1 Automatic Sprinkler Protection

Approved fire suppression technical labor will be required to disconnect and connect fire pump, controllers, and accessories. Acceptance Testing is required if any modifications are made to the system.

8.2.2 Fire Alarm / Mass Notification

Approved fire alarm technical labor will be required to disconnect and connect fire alarm, transceiver equipment, and accessories. Acceptance Testing is required if any modifications are made to the system.

8.2.3 Impairment Coordination

The temporary removal and installation of sprinkler heads will require that the existing sprinkler systems be taken out of service. The impairment procedures detailed in UFC 3-601-02, Chapters 1-10 and 1-11 shall apply to all fire protection impairments. Careful impairment handling includes planning of work to limit impairments to a minimal time period. As sprinkler heads are removed for the day's work, caps shall be installed on the open pipe ends to allow the sprinkler system to be returned to service. Building occupants shall be notified of fire protection impairments. Warning signs shall be placed at all principle employee and visitor entrances to the building during the impairment period. In certain cases the cessation of hazardous operations and the establishment of fire watch may be required.

8.3 Means of Egress

8.3.1 Illumination of Means of Egress

Emergency lighting will be provided in accordance with NFPA 101 within the work area. Emergency lighting for remaining building areas is comprised of light fixtures with emergency ballasts and wall mounted emergency lighting and will remain as is.

8.3.2 Egress Travel Distance

The renovation will not alter egress and travel distances. The renovation work will not make the building less conforming with UFC 3-600-01 and NFPA 101 than before the renovation.

8.3.3 Egress Capacity

The renovation will not alter the building egress components. The renovation work will not make the building less conforming with UFC 3-600-01 and NFPA 101 than before the renovation.

8.4 Safeguards During Construction

All work associated with this project shall be performed in accordance with NFPA 241 Standard for Safeguarding Construction, Alteration, and Demolition Operations.

8.4.1 Hot Work

All Hot Work associated with demolition and construction shall be performed in accordance with NFPA 51. Safeguard for Hot Work operations include, but are not limited to the relocation of combustible storage, the use of fire resistive barriers to protect combustibles that cannot be relocated, the establishment of Fire Watch and the termination of hot work operations when fire protection systems are out of service. Additionally, Hot Work should not be performed when alternative methods can be employed to dismantle existing or assemble new systems.

8.4.2 Fire Protection Impairment Supervision

Fire protection impairments will be supervised and performed in accordance with UFC 3-601-02.



Appendix A

Acronyms and Abbreviations

A	Ampere(s)
AABC	Associated Air Balance Council
ACCA	Air Conditioning Contractors Association
ACI	American Concrete Institute
ACS	Access Control System
ACT	Acoustic Ceiling Tile
AFB	Air Force Base
AFF	Above Finished Floor
AFI	Air Force Instruction
AHJ	Authorities Having Jurisdiction
AHU	Air Handling Unit
AISC	American Institute of Steel Construction
ANSI	American National Standards Institute
ASCE	American Society of Civil Engineers
ASHRAE	American Society of Heating, Refrigeration and Air-Conditioning Engineers
ASTM	American Society for Testing and Materials
BTHU	British thermal unit
BUR	Built-Up Roof
CAC	Common Access Card
CBT	Computer Based Training
CCTV	Closed Circuit Television
CIPS	Cyberspace Infrastructure Planning System
CMU	Concrete Masonry Unit
COMM	Communications
DB	Dry Bulb
DoD	Department of Defense
DX	Direct Expansion
E	East
EBI	Enterprise Building Integrator
EMI	Electromagnetic Interference
EMT	Electrical Metallic Tubing
FCI	Fire Control Instruments
FCU	Fan Coil Unit

FF-L	Federal Specification
FPS	Feet Per Second
FT	Feet
GSA	General Services Administration
HR	Hour(s)
HSS	Hollow Structural Section
HVAC	Heating, Ventilation and Air Conditioning
IAW	In Accordance With
IBC	International Building Code
IC	Intelligence Community
ICD	Intelligence Community Directive
ICS	Intelligence Community Standard
IDS	Intrusion Detection System
IEBC	International Existing Building Code
IECC	International Energy Conservation Code
IMC	International Mechanical Code
IN	Inches
IPC	International Plumbing Code
IT	Information Technology
kVA	Kilovolt-Ampere
LAN	Local Area Network
LED	Light Emitting Diode
MOD	Motorized Damper
N	North
NEBB	National Environmental Balancing Bureau
NFPA	National Fire Protection Association
NIPR	Non-Secure Internet Protocol Router
PSF	Pounds per Square Foot
RF	Radio Frequency
RFI	Radio-Frequency Interference
SIPR	Secure Internet Protocol Router
SMACNA	Sheet Metal and Air Conditioning Contractors' National Association
S	South

STC	Sound Transmission Class
TAFB	Tinker Air Force Base
UFC	Unified Facilities Criteria
UPS	Uninterruptible Power Supply
V	Volts
VAV	Variable Air Volume
W	West
WB	Wet Bulb
WG	Water Gauge
§	Subsection
°F	Fahrenheit



Appendix B

Structural Calculations and Cutsheets



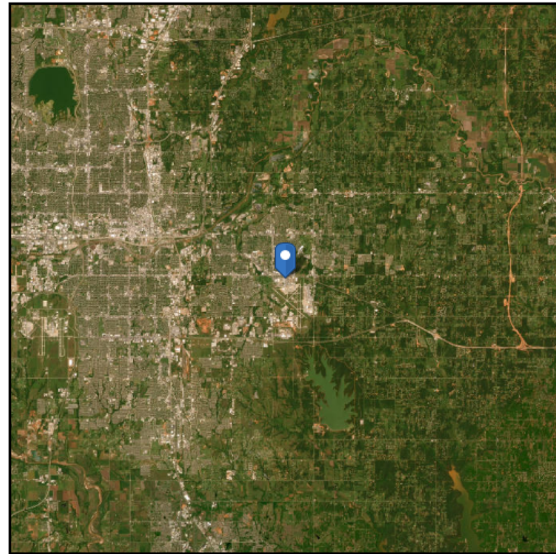
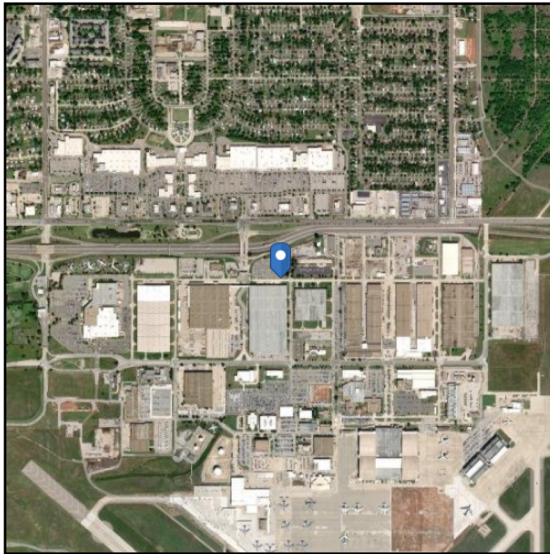


Address:
7535 5th St
Tinker AFB, Oklahoma
73145

ASCE Hazards Report

Standard: ASCE/SEI 7-22
Risk Category: IV
Soil Class: D - Stiff Soil

Latitude: 35.433247
Longitude: -97.395859
Elevation: 1238.948531260525 ft
(NAVD 88)



Wind

Results:

Wind Speed	120 Vmph
10-year MRI	75 Vmph
25-year MRI	82 Vmph
50-year MRI	88 Vmph
100-year MRI	93 Vmph
300-year MRI	102 Vmph
700-year MRI	109 Vmph
1,700-year MRI	115 Vmph
3,000-year MRI	120 Vmph
10,000-year MRI	130 Vmph
100,000-year MRI	149 Vmph
1,000,000-year MRI	168 Vmph

Data Source:

Date Accessed:

ASCE/SEI 7-22, Fig. 26.5-1D and Figs. CC.2-1-CC.2-4,
and Section 26.5.2
Mon Mar 24 2025



Value provided is 3-second gust wind speeds at 33 ft above ground for Exposure C Category, based on linear interpolation between contours. Wind speeds are interpolated in accordance with the 7-22 Standard. Wind speeds correspond to approximately a 1.6% probability of exceedance in 50 years (annual exceedance probability = 0.00033, MRI = 3,000 years). Values for 10-year MRI, 25-year MRI, 50-year MRI and 100-year MRI are Service Level wind speeds, all other wind speeds are Ultimate wind speeds.

Site is not in a hurricane-prone region as defined in ASCE/SEI 7-22 Section 26.2.

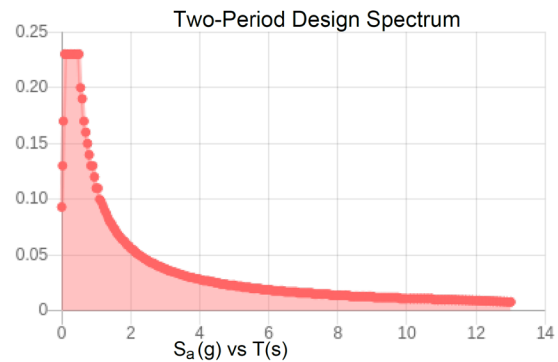
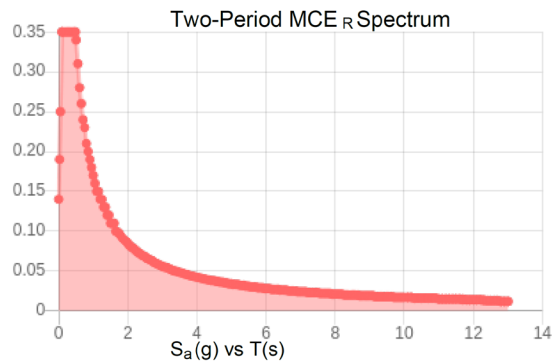
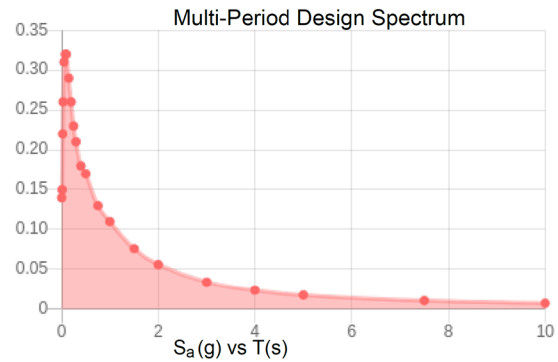
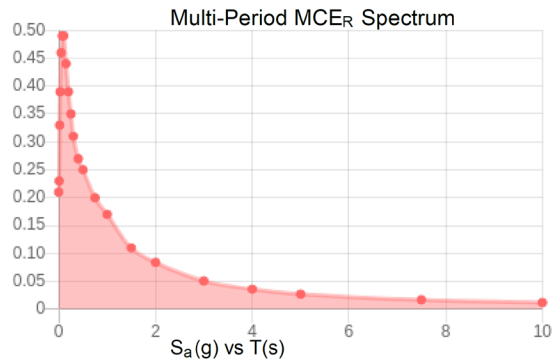
Site Soil Class:

D - Stiff Soil

Results:

PGA_M :	0.19	T_L :	12
S_{MS} :	0.35	S_S :	0.31
S_{M1} :	0.17	S_1 :	0.08
S_{DS} :	0.23	V_{S30} :	260
S_{D1} :	0.11		

Seismic Design Category: C



MCE_R Vertical Response Spectrum
Vertical ground motion data has not yet been made available by USGS.

Design Vertical Response Spectrum
Vertical ground motion data has not yet been made available by USGS.



Data Accessed:

Mon Mar 24 2025

Date Source:

USGS Seismic Design Maps based on ASCE/SEI 7-22 and ASCE/SEI 7-22 Table 1.5-2. Additional data for site-specific ground motion procedures in accordance with ASCE/SEI 7-22 Ch. 21 are available from USGS.

Ice

Results:

Ice Thickness:	1.95 in.
Concurrent Temperature:	5 F
3-s Gust Speed	40 mph

Data Source: Standard ASCE/SEI 7-22, Figs. 10-2 through 10-8

Date Accessed: Mon Mar 24 2025

Ice thicknesses on structures in exposed locations at elevations higher than the surrounding terrain and in valleys and gorges may exceed the mapped values.

Values provided are equivalent radial ice thicknesses due to freezing rain for 250, 500, 1,000, and 1,400-year mean recurrence intervals along with concurrent 3-s gust speeds and concurrent air temperatures. The shading indicates special icing regions, with elevations above 2,100 ft (640 m) in the east, 6,000 ft (1829 m) in the west, and 1,600 ft (488 m) in Alaska, with sparse weather station data for determining design ice loads. In these regions, as well as in regions with complex terrain causing unusual icing conditions and regions where snow or in-cloud icing results in larger loads, the mapped values should be adjusted based on a combination of local historical records and experience, reanalysis data, and numerical weather prediction systems.

Snow

Results:

Ground Snow Load, p_g :	22 lb/ft ²
20-year MRI Value:	7.13 lb/ft ²
Winter Wind Parameter:	0.55
Mapped Elevation:	1232.1 ft

Data Source: ASCE/SEI 7-22, Figures 7.6-1 and 7.6-2 A-D

Date Accessed: Mon Mar 24 2025

Values provided are ground snow loads. In areas designated "case study required," extreme local variations in ground snow loads preclude mapping at this scale. Site-specific case studies are required to establish ground snow loads at elevations not covered.

Snow load values are mapped to a 0.5 mile resolution. This resolution can create a mismatch between the mapped elevation and the site-specific elevation in topographically complex areas. Engineers should consult the local authority having jurisdiction in locations where the reported 'elevation' and 'mapped elevation' differ significantly from each other.

Ground Snow Loads for IRC only, $p_{g(asd)}$: 15.4 lb/ft²



Rain

Results:

15-minute Precipitation Intensity: 9.84 in./h

60-minute Precipitation Intensity: 5.06 in./h

Data Source:

NOAA National Weather Service, Precipitation
Frequency Data Server, Atlas 14([https://](https://www.nws.noaa.gov/oh/hdsc/)

Date Accessed:

www.nws.noaa.gov/oh/hdsc/
Mon Mar 24 2025

Results:

Flood Zone Categorization:

X (unshaded)

Base Flood Elevation:

Refer to map for local elevations and interpolate according to the Authority Having Jurisdiction.

Data Source:

FEMA National Flood Hazard Layer - Effective Flood Hazard Layer for US, where modernized (<https://msc.fema.gov/portal/search>)
Mon Mar 24 2025

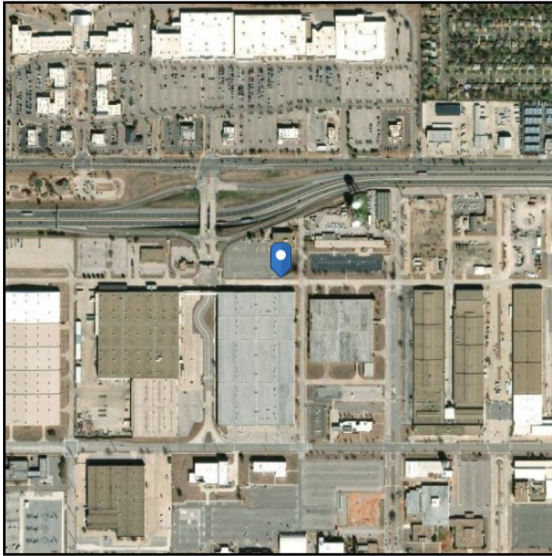
Date Accessed:

FIRM Panel:

If available, download FIRM panel [here](#)

Insurance Study Note:

Download FEMA Flood Insurance Study for this area [here](#)





Tornado

Results:

RC = IV
(MRI =
3,000
years)

MRI =	MRI =	MRI =	MRI =
10,000	100,000	1,000,000	10,000,000
years	years	years	years

Effective Plan Area (ft ²)	Tornado Speed (mph)	Tornado Speed (mph)	Tornado Speed (mph)	Tornado Speed (mph)	Tornado Speed (mph)
A _e = 1	V _T = 95	V _T = 123	V _T = 174	V _T = 220	V _T = 256
A _e = 2,000	V _T = 96	V _T = 125	V _T = 175	V _T = 222	V _T = 259
A _e = 10,000	V _T = 99	V _T = 128	V _T = 177	V _T = 223	V _T = 261
A _e = 40,000	V _T = 103	V _T = 132	V _T = 183	V _T = 226	V _T = 265
A _e = 100,000	V _T = 107	V _T = 136	V _T = 185	V _T = 230	V _T = 267
A _e = 250,000	V _T = 113	V _T = 142	V _T = 191	V _T = 234	V _T = 270
A _e = 1,000,000	V _T = 125	V _T = 153	V _T = 200	V _T = 241	V _T = 277
A _e = 4,000,000	V _T = 138	V _T = 164	V _T = 211	V _T = 251	V _T = 286

To select the appropriate tornado hazard map, the effective plan area, A_e, of the building, other structure, or facility, shall be determined in accordance with Section 32.5.4 and shall be rounded up to the next available mapped A_e. Alternatively, linear interpolation of tornado speed between maps using the logarithm of the effective plan area size is permitted, per Section 32.5.1.

<https://ascehazardtool.org/>

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Data Source:

[ASCE Tornado Geodatabase](#)

Date Accessed:

Mon Mar 24 2025

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